ENGINEERING ECONOMICS

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JESSE WENCIE R. NIDEA, ECE

PROBLEM #1

INTEREST
Present Value

$$F = P(1 + i)^n \rightarrow P = F\,(1 + i)^{-n}$$

Single Payment Present Worth Factor (SPPWF)

$$(1 + i)^{-n} = \frac{1}{(1+i)^n} = \left(\frac{P}{F}, i\,\%, n\right)$$

Quarterly deposits were made in an investment fund earning 15% quarterly for 2 years. The initial deposit was $2,000 and each succeeding deposits, hence, was $250 more than the previous. Determine the fund at the end of the 1st and 2nd year. Draw the CFD. Use SPPWF as notation and compute the amounts.

PROBLEM #2

$$(1 + i)^{-n} = \frac{1}{(1+i)^n} = \left(\frac{P}{F}, i\,\%, n\right)$$

Solution_NewCar

INTEREST
Present Value

I intend to retire my SUV at the end of 2027 to replace with a Toyota Fortuner at an estimated cost of ₱2.5M. My car is expected to be sold for ₱0.55M upon replacement. To provide for the replacement, my employer intends to deposit the following amounts in a vehicle fund that earns an interest at 7.25% QI:

- ₱250,000 at Q1 of 2025
- ₱350,000 at Q3 of 2025
- ₱175,000 at Q2 of 2026
- ₱228,000 at Q3 of 2026

How much is the additional amount needed at the end of 2027 to purchase my dream car?

Make the CFD and use the SPPWF as notation to compute the final amount?

PROBLEM #2

INTEREST
Present Value

$$(1 + i)^{-n} = \frac{1}{(1+i)^n} = \left(\frac{P}{F}, i\,\%, n\right)$$

Somebody owe somebody ₱500,000 with a payment schedule as follow:

Payment Sked	Time	Timeline	Rate of Interest
1	₱150,000	eof 2 months	6.75% compounded monthly
2	₱30,000	eof 8 months	
3	₱120,000	eof 12 months	
4	₱85,000	eof of 14 months	
5 (Finale)	?	eof 24 month	

Find the finale payment. Make the CFD and use the SPPWF as a notation to compute the finale amount?

Finale_Payment

PROBLEM #2

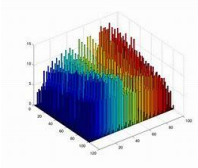
$$(1 + i)^{-n} = \frac{1}{(1+i)^n} = \left(\frac{P}{F}, i\,\%, n\right)$$

Trust

INTEREST
Present Value

Annual deposits of $35,000 were made for 3 years in a trust fund that earn 8% continuous compounding for the first year (**$F=Pe^{rn}$**). Year 2 was 5% compounded semi-annually while, Year 3 was 12% compounded annually. The last 2 years is 20% and 60% less of the initial amount, respectively.

- Determine the amount in the trust fund after Year 3.
- Make the CFD and use the SPPWF as a notation to compute the required amount?
- Make a graph to show the trend of payment.



Interest, Money, Time & Relationship

1. ANNUITY AND CAPITALIZED COST

ANNUITY (uh-noo-uh-tee)

- Is a financial product that provides a series of payments made at equal intervals.
- Used in investments, loans and retirement planning

Examples:

- ✓ Monthly savings deposit
- ✓ Fixed pension payments

Importance:

- Helps individuals plan for financial stability in the future.
- Used by businesses, insurance companies and individuals.

ORDINARY ANNUITY

$$FV = \frac{A[(1+r)^n - 1]}{r} \qquad PV = \frac{A[1-(1+r)^{-n}]}{r}$$

- **Present Value (PV)** = the current worth of future annuity payments.
- **Future Value (FV)** = the total value accumulated over time
- **Annuity (A)** = the total amount paid per period
- **r** = interest rate
- **n** =total number of payments

SAMPLE PROBLEM:

What are the present worth and the accumulated amount of a 5 year annuity paying ₱5,000 at the end of each year, with a 13% interest compounded semi-annually? Draw the CFD and solve.

Interest, Money, Time & Relationship

2. ORDINARY ANNUITY vs. ANNUITY DUE

Feature	Ordinary Annuity	Annuity Due
Payment Timing	End of period	Beginning of Period
Interest Accumulation	Less due to delayed payments	More due to earlier payments
Formula	$FV = A * \frac{(1+r)^n - 1}{r}$	$FV_{due} = FV * (1+r)$

- **Ordinary Annuity:** Rent payments made at the end of the month.
- **Annuity Due:** Lease payments made at the start of the month.

Interest, Money, Time & Relationship

3. TYPES OF ANNUITY

ANNUITY (uh-noo-uh-tee)

- **Ordinary Annuity** – Payments at end of period
- **Annuity Due** – Payments at beginning of period
- **Deferred Annuity** – Starts after delay
- **Perpetuity** – Infinite payments
- **Fixed Annuity** – Constant payments
- **Variable Annuity** – Payments vary with returns
- ***Arithmetic Gradient Series***

$$FV_{\text{gradient}} = G \cdot \left[\left(\frac{(1+i)^n - 1}{i} \right) - \frac{n}{(1+i)^n} \right] \cdot \left(\frac{1}{i} \right)$$

$$PV_{\text{gradient}} = G \cdot \left(\frac{1}{i} \cdot \left[\frac{1 - \frac{n}{(1+i)^n}}{i} \right] \right)$$

4. Problem Sets

Draw the Cash Flow Diagram and Solve:

1. Project A: $18,000/year for 6 years compounded quarterly at 7% nominal.
   Project B: $15,000/year + $3,000 reinvested quarterly. Which is better?

2. A firm invests $10,000/year at 6.5% **yearly**. How long to reach $90,000?

3. Engineer wants $120,000 in 6 years from $15,000/year reinvested semi-annually. What rate is required?

4. Project A: $20k/year for 5 years now.
   Project B: $25k/year starting year 4. Compare at 9% quarterly.

5. **Reinvestment Rate Required**
   Project A: $15,000/year for 6 years, 8% semi-annual
   Project B: $12,000/year + $3,000 reinvested/year
   Find reinvestment rate so Project B equals Project A.

6. **Duration for Equal Value**
   Project A: $10,000/year reinvested at 8%
   Project B: $12,000/year for 6 years
   Find how long A must run to match B.

5. Problem Sets

Draw the Cash Flow Diagram and Solve:

7. **Required Annuity for Equipment**
   Target: $60,000 in 5 years
   Find annuity (X) at 7%
   Compare with Project B paying $13,000/year

8. **Semi-Annual Compounding Decision**
   A: $18,000/year, 6 years @ 6% semi-annually
   B: $15,000/year + $3,000 reinvested semi-annually

9. **Quarterly Compounding Duration**
   Invest $10,000/year with 7% quarterly compounding
   Target = $90,000

6. Problem Sets

Draw the Cash Flow Diagram and Solve:

11. Bi-Monthly Reinvestment Rate

$15,000/year invested to reach $120,000 in 6 years

Bi-monthly compounding: what rate required?

12. Deferred Annuity with Quarterly Compounding

A: 10K initially and $20k/year after 3 year;

B: $25k/year starting year 4. time period is 6 years.

Compare at 9% quarterly compounding